

Golden-ratio polylogarithm ladders: reconstruction, extension to weight nine, and Reshetnikov's conjectures

Vladimir Reshetnikov

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Abstract

A polylogarithm ladder packages, for a fixed algebraic base and a range of weights n , a rational-coefficient combination $\sum_k c_k \operatorname{Li}_n(\rho^{a_k})$ that collapses to $\zeta(n)$ plus π -powers times logarithms. For the golden ratio ϕ (with $\rho = \phi^{-1}$, $1 - \rho^2 = \rho$) these were charted by Abouzahra and Lewin (1985) and revisited on MathOverflow by Reshetnikov and Tito Piezas III. We verify Reshetnikov's tetralogarithm conjecture (MSE 1373123) and his three Li_5 golden conjectures (MO 261408) to several hundred digits, confirm Tito's golden family at weights 2–5, and—this is the new part—*reconstruct the explicit weight-6 and weight-7 coefficients Tito found but left unwritten* (“too tedious to write here”), with the $\zeta(6)$ and $\zeta(7)$ coefficients coming out to exactly his stated 32155 and 44689995. We then push the family two weights further than the published material: explicit *weight-8 and weight-9* golden ladders, both on the eight-argument union set $\{\phi^{-2}, \phi^{-4}, \phi^{-6}, \phi^{-8}, \phi^{-10}, \phi^{-12}, \phi^{-20}, \phi^{-24}\}$. The six-argument family caps at weight 7; reaching weights 8–9 requires the larger union of the index-12, 20, 24 argument sets—so the family does not stop at weight 7, the argument set *grows*.

1 Setup

With $\phi = \frac{1+\sqrt{5}}{2}$, $\rho = \phi^{-1}$, the defining relation $1 - \rho^2 = \rho$ is the seed of the golden ladders. Reshetnikov's conjectures live among the values $\operatorname{Li}_n(\phi^{-2k})$; all coefficients below were obtained or verified by PSLQ (`mpmath`, 130–420 digits, Champernowne canary) and, where the cancellation depth allows, cross-checked in Wolfram.

Remark 1. These golden combinations cancel $\sim 10^8$ -sized terms down to zero across 100+ digits. Wolfram's adaptive N silently under-resolves this even at large `$MaxExtraPrecision`; `mpmath`'s explicit precision tracking is authoritative here. (Per-term polylogarithm values agree between the two engines.)

2 Reshetnikov's conjectures, verified

The tetralogarithm conjecture (MSE 1373123, MO 261408) holds: with $\alpha = \ln 2$, $\beta = \ln 3$,

$$\begin{aligned} 720 \operatorname{Li}_4\left(\frac{1}{2}\right) - 2160 \operatorname{Li}_4\left(\frac{1}{3}\right) + 2160 \operatorname{Li}_4\left(\frac{2}{3}\right) + 270 \operatorname{Li}_4\left(\frac{1}{4}\right) + 540 \operatorname{Li}_4\left(\frac{3}{4}\right) + 135 \operatorname{Li}_4\left(\frac{1}{9}\right) \\ = 19\pi^4 + 30(\pi^2 - \beta^2)(10\alpha^2 - 12\alpha\beta + 3\beta^2) - 30\alpha^2(19\alpha^2 - 24\alpha\beta + 8\beta^2), \end{aligned} \quad (1)$$

and is in fact the *unique* integer relation (height 2160) among those six tetralogarithms and the weight-4 elementary basis $\{\pi^4, \pi^2 \ln^2, \ln^4, \zeta(3) \ln\}$ —a rational base-2×base-3 ladder. The three Li_5 golden conjectures also hold (verified to $\sim 10^{-257}$):

$$\begin{aligned} 4455 \operatorname{Li}_5(\phi^{-2}) - 1215 \operatorname{Li}_5(\phi^{-4}) - 360 \operatorname{Li}_5(\phi^{-6}) + 15 \operatorname{Li}_5(\phi^{-12}) &= 3015\zeta(5) - 702 \ln \phi^5 + 180\pi^2 \ln \phi^3 - 38\pi^4 \ln \phi, \\ 45000 \operatorname{Li}_5(\phi^{-2}) - 16875 \operatorname{Li}_5(\phi^{-4}) - 144 \operatorname{Li}_5(\phi^{-10}) + 9 \operatorname{Li}_5(\phi^{-20}) &= 28944\zeta(5) - 6000 \ln \phi^5 + 1600\pi^2 \ln \phi^3 - 356\pi^4 \ln \phi, \\ 15660 \operatorname{Li}_5(\phi^{-2}) - 19440 \operatorname{Li}_5(\phi^{-4}) + 7680 \operatorname{Li}_5(\phi^{-6}) + 2430 \operatorname{Li}_5(\phi^{-8}) - 15 \operatorname{Li}_5(\phi^{-24}) &= 5025\zeta(5) + 2088 \ln \phi^5 - 240\pi^2 \ln \phi^3 - 28\pi^4 \ln \phi. \end{aligned}$$

3 Completing Tito's family: the unwritten weight-6 and weight-7 ladders

Tito's "old answer" lists the golden family at weights 2–5 (all verified here), then states the weight-6 and weight-7 ladders only schematically, the integer coefficients "too tedious to write." We recover them by PSLQ. On the six-argument set $\{\phi^{-2}, \phi^{-4}, \phi^{-6}, \phi^{-8}, \phi^{-12}, \phi^{-24}\}$:

$$\begin{aligned} & 98820 \operatorname{Li}_6(\phi^{-2}) - 104490 \operatorname{Li}_6(\phi^{-4}) + 33120 \operatorname{Li}_6(\phi^{-6}) + 7290 \operatorname{Li}_6(\phi^{-8}) - 50 \operatorname{Li}_6(\phi^{-12}) - 15 \operatorname{Li}_6(\phi^{-24}) \\ &= -13032 \ln \phi^6 + 3240\pi^2 \ln \phi^4 - 424\pi^4 \ln \phi^2 + \mathbf{32155} \zeta(6), \end{aligned} \quad (2)$$

$$\begin{aligned} & 74707920 \operatorname{Li}_7(\phi^{-2}) - 39497220 \operatorname{Li}_7(\phi^{-4}) + 8346240 \operatorname{Li}_7(\phi^{-6}) + 1377810 \operatorname{Li}_7(\phi^{-8}) - 6300 \operatorname{Li}_7(\phi^{-12}) - 945 \operatorname{Li}_7(\phi^{-24}) \\ &= 2814912 \ln \phi^7 - 979776\pi^2 \ln \phi^5 + 213696\pi^4 \ln \phi^3 - 51448\pi^6 \ln \phi + \mathbf{44689995} \zeta(7). \end{aligned} \quad (3)$$

The $\zeta(6)$ and $\zeta(7)$ coefficients reproduce Tito's stated 32155 and 44689995 *exactly*—an independent confirmation that the reconstructions (2)–(3) are correct (residuals 10^{-117} , 10^{-114} by direct substitution; coefficients primitive).

4 Two weights beyond: explicit ladders at weight 8 and 9

Tito remarks that the index-24 ladder "can be extended even to $n = 9$ " but gives no formula. We find the explicit ladders. The six-argument set of (2)–(3) closes at weights 2–7 but *not* at 8 or 9. Reaching them requires the *eight*-argument union of the index-12, 20, 24 sets, $\mathcal{A} = \{\phi^{-2}, \phi^{-4}, \phi^{-6}, \phi^{-8}, \phi^{-10}, \phi^{-12}, \phi^{-20}, \phi^{-24}\}$:

$$\begin{aligned} & -490795200 \operatorname{Li}_8(\phi^{-2}) + 636131475 \operatorname{Li}_8(\phi^{-4}) - 388684800 \operatorname{Li}_8(\phi^{-6}) + 82668600 \operatorname{Li}_8(\phi^{-8}) \\ &+ 9507456 \operatorname{Li}_8(\phi^{-10}) + 5250000 \operatorname{Li}_8(\phi^{-12}) - 74277 \operatorname{Li}_8(\phi^{-20}) - 18900 \operatorname{Li}_8(\phi^{-24}) \\ &= -23641920 \ln \phi^8 + 7956480\pi^2 \ln \phi^6 - 1434720\pi^4 \ln \phi^4 + 184432\pi^6 \ln \phi^2 - 148565886 \zeta(8), \end{aligned} \quad (4)$$

$$\begin{aligned} & 220857840000 \operatorname{Li}_9(\phi^{-2}) - 143129581875 \operatorname{Li}_9(\phi^{-4}) + 58302720000 \operatorname{Li}_9(\phi^{-6}) - 9300217500 \operatorname{Li}_9(\phi^{-8}) \\ &- 855671040 \operatorname{Li}_9(\phi^{-10}) - 393750000 \operatorname{Li}_9(\phi^{-12}) + 3342465 \operatorname{Li}_9(\phi^{-20}) + 708750 \operatorname{Li}_9(\phi^{-24}) \\ &= -2364192000 \ln \phi^9 + 1022976000\pi^2 \ln \phi^7 - 258249600\pi^4 \ln \phi^5 + 55329600\pi^6 \ln \phi^3 - 14149132\pi^8 \ln \phi + 125596001790 \zeta(9). \end{aligned} \quad (5)$$

Both are verified by direct substitution (residuals 0 at 150 digits for (4), 10^{-190} at 200 digits for (5)); coefficients are primitive. The weight-9 ladder appeared identically across three independent searches with different padding arguments (the extra arguments returned coefficient 0), a strong guard against over-fitting.

Remark 2. So the golden family does *not* cap at weight 7: the closure persists, but the argument set grows—four or five arguments per index at weight 5, six arguments at weights 6–7, eight arguments (the full union) at weights 8–9. Whether the growth can be sustained to arbitrarily high weight is exactly the question Abouzahra–Lewin address in the negative; our data are consistent with an eventual stop once the required argument set outgrows the available golden multiplicative relations.

5 The proof of Reshetnikov's tetralogarithm conjecture

The conjecture (1) is provable from elementary functional equations. The six tetralogarithms span a 5-dimensional space modulo the weight-4 elementary basis ($\operatorname{Li}_4(\frac{1}{9})$ is the only reducible member), so (1) is the unique relation among them. Two duplications $\operatorname{Li}_4(\frac{1}{4}) = 8(\operatorname{Li}_4(\frac{1}{2}) + \operatorname{Li}_4(-\frac{1}{2}))$ and $\operatorname{Li}_4(\frac{1}{9}) = 8(\operatorname{Li}_4(\frac{1}{3}) + \operatorname{Li}_4(-\frac{1}{3}))$ reduce the left side to the reflected arguments, and a single *cross-base Kummer relation* then cancels every tetralogarithm:

$$\begin{aligned} \operatorname{Li}_4(\tfrac{3}{4}) + 4 \operatorname{Li}_4(\tfrac{2}{3}) &= -\tfrac{16}{3} \operatorname{Li}_4(\tfrac{1}{2}) - 4 \operatorname{Li}_4(-\tfrac{1}{2}) + 2 \operatorname{Li}_4(\tfrac{1}{3}) - 2 \operatorname{Li}_4(-\tfrac{1}{3}) + \tfrac{19}{540} \pi^4 + \tfrac{5}{9} \pi^2 \ln^2 2 + \tfrac{1}{6} \pi^2 \ln^2 3 \\ &\quad - \tfrac{2}{3} \pi^2 \ln 2 \ln 3 - \tfrac{19}{18} \ln^4 2 - \tfrac{1}{6} \ln^4 3 + \tfrac{4}{3} \ln^3 2 \ln 3 - \ln^2 2 \ln^2 3 + \tfrac{2}{3} \ln 2 \ln^3 3. \end{aligned} \quad (6)$$

(Verified to residual 0.) Notably, Bailey's base-2 identity is *not* used—its arguments $\operatorname{Li}_4(\pm \frac{1}{8})$ never appear—and there is *no* base-3 analogue of Bailey's identity (a search over $\{\operatorname{Li}_4(3^{-k})\}$ finds no relation in $\{\pi, \ln 3\}$). The base-2/base-3 asymmetry is genuine.

6 Canonical index-20/24 ladders and a ζ -elimination chain

Reshetnikov's Li_5 conjectures are the weight-5 entries of three first-degree golden ladders (Abouzahra–Lewin). In canonical (monic, a^{n-1} -normalized) form, with $\rho = \phi^{-1}$, $\ell = \ln \rho$, and $\operatorname{tw}(n) = D_1 \frac{\ell^n}{n!} + D_2 \zeta(2) \frac{\ell^{n-2}}{(n-2)!} +$

$$D_3 \zeta(4) \frac{\ell^{n-4}}{(n-4)!}:$$

$$L_{20}(n) = \frac{\text{Li}_n(\rho^{20})}{20^{n-1}} - \frac{\text{Li}_n(\rho^{10})}{10^{n-1}} - 3 \frac{\text{Li}_n(\rho^4)}{4^{n-1}} + \frac{1}{2} \frac{\text{Li}_n(\rho^2)}{2^{n-1}} + \text{tw}(n), \quad (D_1, D_2, D_3) = (-\frac{1}{2}, \frac{1}{25}, -\frac{89}{4000}), \quad L_{20}(5) = \frac{201}{10000} \zeta(5),$$

$$L_{24}(n) = \frac{\text{Li}_n(\rho^{24})}{24^{n-1}} - \frac{4}{3} \frac{\text{Li}_n(\rho^{12})}{12^{n-1}} - 2 \frac{\text{Li}_n(\rho^8)}{8^{n-1}} + \frac{7}{3} \frac{\text{Li}_n(\rho^4)}{4^{n-1}} - \frac{205}{576} \frac{\text{Li}_n(\rho^2)}{2^{n-1}} + \text{tw}(n), \quad (D_1, D_2, D_3) = (\frac{179}{576}, -\frac{5}{192}, \frac{629}{41472}), \quad L_{24}(5) = -\frac{1541}{110592} \zeta(5),$$

each vanishing for $n \leq 4$. The shared factor 67 in $201 = 3 \cdot 67$, $1541 = 23 \cdot 67$ is what lets the three indices jointly eliminate $\zeta(5)$. Combining them removes higher zetas: $M_{20} = L_{20} - \frac{1296}{625} L_{12}$ and $M_{24} = L_{24} + \frac{23}{16} L_{12}$ kill $\zeta(5)$ and reach weight 7, and $T = M_{24} + \frac{409375}{373248} M_{20}$ kills $\zeta(7)$ and reaches weight 9, with $T(9) = \frac{66452911}{41278242816000} \zeta(9) + \frac{3537283}{2063912140800} \zeta(8) \ell - \frac{11527}{2866544640} \zeta(6) \frac{\ell^3}{3!}$ (confirmed by direct substitution and an independent PSLQ at 460 digits). This is the Abouzahra–Lewin route to weight 9, distinct from the single eight-argument ladders (4)–(5).

7 A new ladder base: the minimal Pisot number

Beyond the golden ratio, the minimal Pisot number $\omega = 0.7548776\dots$ (the real root of $x^3 + x^2 - 1$, the base Abouzahra–Lewin deferred to a never-published sequel) carries its own ladder. Its seeds are pure units— $1 - \omega = \omega^5$, $1 - \omega^2 = \omega^3$, $1 - \omega^3 = \omega^2$, $1 - \omega^5 = \omega$ —and the first-degree ladder climbs to weight 4 (with $\lambda = \ln \omega$):

$$\begin{aligned} \frac{1}{2} \text{Li}_2(\omega^2) - \text{Li}_2(\omega^5) - \lambda^2 &= 0, & -\text{Li}_3(\omega) + \frac{6}{4} \text{Li}_3(\omega^2) + \frac{3}{9} \text{Li}_3(\omega^3) - \text{Li}_3(\omega^5) - \frac{6\lambda^3}{3!} &= 0, \\ -71 \text{Li}_4(\omega) - \frac{120}{8} \text{Li}_4(\omega^2) + \frac{1053}{27} \text{Li}_4(\omega^3) - 17 \text{Li}_4(\omega^5) + \frac{5488}{343} \text{Li}_4(\omega^7) - \text{Li}_4(\omega^{14}) + 51\zeta(4) + \frac{2010\lambda^4}{4!} - \frac{84\zeta(2)\lambda^2}{2!} &= 0. \end{aligned}$$

The weight-4 step requires the index-7 leg ω^7, ω^{14} (the “higher index unlocks higher weight” signature) and the ladder stops there. Among the bases tested, this is the only genuinely new one: $x^3 + x - 1$, $x^4 + x^3 - 1$, $x^5 + x^4 - 1$, $x^6 + x - 1$ and others support only the trivial weight-2 reflection, and $x^4 + x^2 - 1$ is the golden ratio in disguise.

8 Hunting more ladder bases, and a methodological caveat

Beyond ϕ and ω , which other bases admit ladders? A pure-unit seed is a relation $1 - \rho^a = \rho^b$; ladders need several. A brute-force scan over *all* degree- ≤ 5 monic unit polynomials finds that ω (the minimal Pisot number) is the *unique* base with ≥ 3 pure-unit seeds—it has four, $1 - \omega = \omega^5$, $1 - \omega^2 = \omega^3$, $1 - \omega^3 = \omega^2$, $1 - \omega^5 = \omega$. The golden ratio and the supergolden reciprocal $x^3 + x - 1$ have two; the only other “four-seed” bases, $x^6 + x^2 - 1$ and $x^6 + x^3 - 1$, are *root-disguises* (ρ^2, ρ^3 land on the supergolden/golden reciprocals), as $x^4 + x^2 - 1$ is $\sqrt{\text{golden}}$. The n -bonacci reciprocals and the smallest quartic Salem number have ≤ 1 seed and support no ladder.

Two bases the earlier survey had dismissed as “weight-2 only” do, in fact, carry weight-3 ladders. With ρ the root of $x^3 + x - 1$ (the supergolden reciprocal) and of $x^4 + x - 1$ (the reciprocal of the second-smallest Pisot number $x^4 - x^3 - 1$),

$$\begin{aligned} -60 \text{Li}_3(\rho) + 54 \text{Li}_3(\rho^2) - 32 \text{Li}_3(\rho^3) - 3 \text{Li}_3(\rho^6) &= 24 \ln^3 \rho - 4\pi^2 \ln \rho - 36\zeta(3), \\ -12 \text{Li}_3(\rho) + 12 \text{Li}_3(\rho^3) - 12 \text{Li}_3(\rho^4) - 3 \text{Li}_3(\rho^6) &= 22 \ln^3 \rho - 2\pi^2 \ln \rho - 12\zeta(3), \end{aligned}$$

each verified in two engines (and by direct substitution to 400+ digits). The lesson is methodological: **a PSLQ “no relation” is not a proof of non-existence.** The first relation above was *missed* by a pure-tower PSLQ search at 300 digits, yet holds by direct substitution to 400—so the survey’s negative verdicts were false negatives. The reliable test is to take a candidate relation and substitute it at precision far above where it was found. Whether these two bases climb past weight 3 is left open here; the question is delicate precisely because PSLQ negatives cannot settle it.

9 Verification and reproducibility

Equations (1)–(5), Tito’s weights 2–5, and Bailey’s base-2 identity $\text{Li}_4(\frac{1}{64}) - 8 \text{Li}_4(\frac{1}{8}) - 54 \text{Li}_4(\frac{1}{4}) + 96 \text{Li}_4(\frac{1}{2}) = 5 \ln^4 2 - 2\pi^2 \ln^2 2 + \frac{4}{9} \pi^4$ are in `src/PolyLog/identities/polylog-ladders.wl`; scripts under `src/PolyLog/tools/scratch/ladders/`. The negative result that the cubic base $\rho^3 + \rho = 1$ supports only the weight-2 reflection (its weight-3 relation being generic duplication) is recorded there too—the golden relation $1 - \rho^2 = \rho$ is what makes the tower climb.